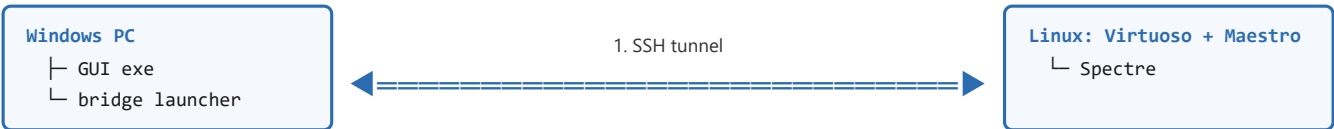


Optimization Tools — User Manual

v0.1.1 · English (default UI language). The GUI also ships a Chinese interface: menu bar **Language** → 中文 at any time.

This manual covers the workflow for optimizing an analog cell with the run release. It assumes you have a Cadence Virtuoso + Maestro environment on a Linux host (physical machine or VM) reachable via SSH from your Windows computer.

0. The workflow at a glance



2. ADE Assemble setup (design variables, outputs, **Single Run**)
3. Optimizer drives & evaluates the iterations
4. Pick a Pareto solution, write back to the schematic

Four stages: **tunnel** → **Cadence-side setup** → **run optimization** → **apply result**.

1. Establish the SSH tunnel + bridge

1. Start your Cadence machine (e.g. VirtualBox VM) and make sure SSH to it works from Windows (`ssh user@host`).
2. Run `start_bridge_launcher/start_bridge_launcher.exe` .
3. Enter the remote host, user and SSH port when prompted (press Enter to accept the shown defaults).
4. The launcher opens the tunnel and starts the bridge service, then prints a `load("../virtuoso_setup.il")` command.
5. Copy that line into the Cadence **CIW** window and press Enter (run `RBStop()` first if a previous session is still loaded).

✓ The tunnel is now established. Closing the launcher window does **not** drop it.

Manual alternative: open the `virtuoso-bridge-lite` project in VS Code and start the tunnel from there. Not needed for daily use.

2. Cadence-side optimization setup

Draw (or open) your schematic; **all parameters to be optimized must be expressed as design variables** (e.g. `w/l/m` referencing global variables), not hard-coded literals.

2.1 ADE Assemble session

Open **ADE Assemble** and, in addition to the usual design/test/corner selection, set up:

#	Tab	What to check
①	Parameters	the design variables to sweep
②	Global Variables	must match the names in the schematic
③	Outputs	the objective measures (see 2.3)
④	Run mode	must be Single Run — with <i>Sweeps & Corners</i> selected the optimizer cannot drive the session

Parameter names must correspond one-to-one with the schematic. You can type them manually, or import them from Cadence's internal optimization support.

2.2 Variables

Add/edit each variable: name, minimum, maximum (and initial value). Once the table looks complete, **save to** `variables.csv` so later runs can reuse it.

2.3 Outputs

Output setup is the usual ADE practice, with two rules for the optimizer:

1. Every row that should be an **optimization objective** must have a `Name` and a `Weight`.
2. Rows **without** a `Name` (raw `expr` outputs, signal traces) are ignored as objectives.

Then export the setup CSV and rename the file:

```
<your_setup>_maestro.csv → <your_setup>_maestro_setup.csv
```

Copy the `..._maestro_setup.csv` file over to the Windows side and point the GUI at it (file picker: output CSV path).

3. Run the optimization

1. Launch `optimization_tool_gui/optimization_tool_gui.exe`.
2. Create or open a project, load `variables.csv` / `specs.csv` / the `..._maestro_setup.csv` you copied in.
3. Choose the algorithm and schedule: single-objective (GA, DE, PSO, CMA-ES, Bayes) or multi-objective (NSGA-II/III, SPEA2, MOEA/D, CTAEA), generations × population; optionally enable the **AI surrogate** (cold-start sampling → predict → periodically verify on Spectre).
4. Press start. The log panel shows each evaluation; convergence / Pareto / history plots update live (double-click a plot to enlarge).
5. You can stop early — completed evaluations are kept in the run ledger, and a later run resumes from the saved state.

4. Apply the result

1. When the run finishes, open the results table (Pareto set for multi-objective runs).
2. Click the solution you want — the GUI writes the parameters back to Cadence.
3. The ADE window refreshes and the schematic reflects the chosen device parameters; run a final confirmation simulation in Cadence as usual.

5. Known issues / notes

- If the automatic export of `..._maestro_setup.csv` does not land on the Windows side, copy it manually from the simulation host (see step 2.3).
- The optimizer drives **one simulation at a time** through the bridge (`parallel_eval=1`); do not touch the ADE session during a run.
- First launch may trigger Windows SmartScreen: *More info* → *Run anyway*.
- Keep each exe folder together (`exe` + `_internal/`); do not move the exe alone.

6. Interface language

Default language is **English**. Switch any time via the menu bar: **Language** → **中文** (菜单「语言」 → English). All dialogs, buttons and log messages translate in place.